

HIGH ACRYLONITRILE NBR: PATENT LANDSCAPE ON ACRYLONITRILE-RICH RUBBER SYNTHESIS AND RELATED COPOLYMER TECHNOLOGIES

1. ABSTRACT

This report surveys a patent set relevant to high-acrylonitrile nitrile butadiene rubber (NBR) and closely related acrylonitrile-containing elastomer/copolymer technologies. The target material is defined as nitrile butadiene rubber with elevated acrylonitrile content, emphasizing monomer ratio control, polymerization route, and molecular-weight/particle-structure management. Within the manifest, the most directly relevant NBR synthesis evidence is limited: several patents address acrylonitrile-containing rubber systems indirectly through ABS graft copolymers, styrene-butadiene binders, or functionalized rubbery polymers rather than dedicated high-ACN NBR recipes. The landscape therefore spans three main technical directions. First, emulsion and graft polymerization approaches dominate the older ABS-related patents, where butadiene rubber latex is grafted with styrene and acrylonitrile, and process variables such as emulsifier loading, initiator level, molecular-weight modifier, electrolyte, staged monomer addition, conversion thresholds, and reaction time are used to control particle formation and impact strength. Second, newer patents focus on application-driven formulations that use nitrile-containing polymers as dispersants or binders in battery electrodes, but these are not NBR synthesis disclosures; they instead highlight the importance of polarity, viscosity, gel content, and glass-transition tuning in acrylonitrile-containing elastomer systems. Third, a controlled emulsion polymerization patent demonstrates a broader methodology for synthesizing block and random copolymers in water using in-situ emulsifier formation and controlled radical techniques, which is conceptually relevant to modern NBR process design even though it is not NBR-specific. Across the set, recurring themes are high-polarity monomer incorporation, control of molecular weight by initiator/chain-transfer strategy, and the use of emulsion or solution polymerization to balance processability with final performance. However, the evidence base here does not provide a direct, parameter-rich high-ACN NBR recipe; instead, it supports the conclusion that high-acrylonitrile NBR synthesis would most plausibly rely on controlled free-radical emulsion or solution polymerization with careful monomer-feed staging and molecular-weight control, while avoiding over-reliance on formulations intended for downstream ABS or battery applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9976027B2
- Patent Title: Polymer fine particle-containing curable resin composition having improved ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples Comparative Example
Composition: - Example Extracted Examples Example Comparative: - Example Extracted
Examples Example Comparative: - Example Extracted Examples Example Comparative: -
Example Extracted Examples Example Comparative: - Example Extracted Examples Example
Comparative: - Example Extracted Examples Example of: - Example Extracted Examples Example
of: - Example Extracted Examples Example of: - Example Extracted Examples example of: -
Example Extracted Examples example of: - Example Extracted Examples example of: - Example
Extracted Examples example of: - Example Extracted Examples example of: - Example Extracted
Examples example of: - Example Extracted Examples example of: - Example Extracted Examples
example of: - Example Extracted Examples Example 1-1: - Example Extracted Examples Example
1-2: - Example Extracted Examples Example 1-1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US11827830B2
- Patent Title: Dispersant, conductive material dispersion and slurry for electrode film
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

- Patent Number: US4559386A

- Patent Title: ABS compositions and process for preparing same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1985
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples EXAMPLE 2--COMPARATIVE:
 - Example Extracted Examples Example 2: - Example Extracted Examples EXAMPLE
 1--CONTROL: - Example Extracted Examples Example illustrates: - Example Extracted Examples
 Example 1: - Example Extracted Examples Example is: - Example Extracted Examples Example 1:
 - Example Extracted Examples Example 1: - Example Extracted Examples Example 2: - Example
 Extracted Examples Example 1: - Example Extracted Examples Example 2: - Example Extracted
 Examples Example 1: - Example Extracted Examples Example 2: - Example Extracted Examples
 Example 1: - Example Extracted Examples EXAMPLE 3: - Example Extracted Examples Example
 is: - Example Extracted Examples Example 2: - Example Extracted Examples Example 3: -
 Example Extracted Examples Example 1: - Example Extracted Examples Example 2:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: US9583768B2
- Patent Title: Negative electrode slurry composition, lithium ion secondary battery negative ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples Comparative Example 1: -
 Example Extracted Examples Comparative Example 2: - Example Extracted Examples
 Comparative Example 3: - Example Extracted Examples Comparative Example 4: - Example
 Extracted Examples Comparative Example 5: - Example Extracted Examples Comparative
 Example 6: - Example Extracted Examples Comparative Example 7: - Example Extracted
 Examples example cokes: - Example Extracted Examples Example 1: - Example Extracted
 Examples Example 2: - Example Extracted Examples Example 1: - Example Extracted Examples

Example 3: - Example Extracted Examples Example 1: - Example Extracted Examples Example 4:
- Example Extracted Examples Example 1: - Example Extracted Examples Example 5: - Example
Extracted Examples Example 1: - Example Extracted Examples Example 6: - Example Extracted
Examples Example 1: - Example Extracted Examples Example 7:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples COMPARATIVE EXAMPLE 9: -
Example Extracted Examples example of: - Example Extracted Examples example better: -
Example Extracted Examples example buffering: - Example Extracted Examples example mineral:
- Example Extracted Examples example succinic: - Example Extracted Examples example ROMP:
- Example Extracted Examples example RAFT: - Example Extracted Examples example cleavage:
- Example Extracted Examples example thermoplastics: - Example Extracted Examples
EXAMPLE 1: - Example Extracted Examples EXAMPLE 2: - Example Extracted Examples
EXAMPLE 3: - Example Extracted Examples EXAMPLE 4: - Example Extracted Examples
Example 2: - Example Extracted Examples Example 2: - Example Extracted Examples EXAMPLE
5: - Example Extracted Examples Example 4: - Example Extracted Examples Example 2: -
Example Extracted Examples Example 3:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: US9461308B2
- Patent Title: Electrode for lithium ion secondary battery, lithium ion secondary battery, ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples Comparative Example 1: - Example Extracted Examples Comparative Example 2: - Example Extracted Examples Comparative Example 3: - Example Extracted Examples Comparative Example 4: - Example Extracted Examples Comparative Example 5: - Example Extracted Examples Comparative Example 6: - Example Extracted Examples Comparative Example 7: - Example Extracted Examples Comparative Example 8: - Example Extracted Examples Comparative Example 7: - Example Extracted Examples Example 1: - Example Extracted Examples Example 2: - Example Extracted Examples Example 1: - Example Extracted Examples Example 1: - Example Extracted Examples Example 3: - Example Extracted Examples Example 1: - Example Extracted Examples Example 4: - Example Extracted Examples Example 1: - Example Extracted Examples Example 5: - Example Extracted Examples Example 1: - Example Extracted Examples Example 6:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: US10344104B2
- Patent Title: Copolymer for polymer blend compatibilizer and resin composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples comparative example was: -
Example Extracted Examples example heat: - Example Extracted Examples Example of: -
Example Extracted Examples Example of: - Example Extracted Examples Example of: - Example
Extracted Examples Example of: - Example Extracted Examples Example of: - Example Extracted
Examples Example of: - Example Extracted Examples Example of: - Example Extracted Examples
Example of: - Example Extracted Examples Example of: - Example Extracted Examples Example
of:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: US10344158B2
- Patent Title: Elastomeric film-forming compositions and articles made from the elastomeric ...
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples example a: - Example Extracted Examples example sulphur: - Example Extracted Examples example a: - Example Extracted Examples example sulphur: - Example Extracted Examples example a: - Example Extracted Examples example only: - Example Extracted Examples Example 1: - Example Extracted Examples Example 2: - Example Extracted Examples Example 3: - Example Extracted Examples Example 3: - Example Extracted Examples example be: - Example Extracted Examples Example of: - Example Extracted Examples example of: - Example Extracted Examples example TMTD: - Example Extracted Examples example a: - Example Extracted Examples example be: - Example Extracted Examples example between: - Example Extracted Examples example involves: - Example Extracted Examples example of: - Example Extracted Examples example the:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US6812307B2
- Patent Title: Functionalized monomers for synthesis of rubbery polymers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2004
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples COMPARATIVE EXAMPLE 10: - Example Extracted Examples COMPARATIVE EXAMPLE 18: - Example Extracted Examples COMPARATIVE EXAMPLE 27: - Example Extracted Examples COMPARATIVE EXAMPLE 35: - Example Extracted Examples COMPARATIVE EXAMPLE 113: - Example Extracted Examples COMPARATIVE EXAMPLE 116: - Example Extracted Examples EXAMPLE 1: - Example Extracted Examples EXAMPLE 2: - Example Extracted Examples Example 1: - Example Extracted Examples Example 1: - Example Extracted Examples EXAMPLE 6: - Example Extracted Examples Example 6: - Example Extracted Examples EXAMPLE 9: - Example Extracted Examples Example 6: - Example Extracted Examples Example No: - Example Extracted Examples EXAMPLE 11: - Example Extracted Examples Example 6: - Example Extracted Examples

Example 11: - Example Extracted Examples EXAMPLE 14: - Example Extracted Examples
Example 6:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US10266615B2 and US4559386A both use acrylonitrile-containing graft/ABS chemistry, but US10266615B2 emphasizes emulsion polymerization with staged monomer addition, acrylate cross-linking agent, emulsifier, initiator, electrolyte, and molecular-weight modifier, whereas US4559386A emphasizes continuous mass polymerization and a deliberate mismatch between surface-grafted and matrix acrylonitrile levels.
- US10266615B2 provides explicit process ranges for diene-based rubber latex preparation: 50 to 75 parts conjugated diene monomer, 1 to 3 parts emulsifier, 0.1 to 0.4 parts initiator, 0.1 to 3 parts electrolyte, 0.1 to 0.5 part molecular-weight modifier, and 65 to 100 parts water; US4559386A does not disclose comparable numeric formulation ranges in the retained text.
- US7592409B2 is the most methodologically modern polymerization disclosure in the set, using in-situ emulsifier formation in emulsion polymerization with a polymerization control agent; it is broader than NBR and does not provide NBR-specific monomer ratios, but it is relevant as a controlled emulsion framework for acrylonitrile-containing elastomers.
- US6812307B2 is solution/anionic rather than emulsion-based: it copolymerizes conjugated diolefin monomers with functionalized monomers in organic solvent at 20 to 100 C, initiated by an anionic initiator and without conventional polar modifiers; this contrasts with the radical emulsion routes used in the ABS-related patents.
- US11827830B2 and US9461308B2 are application-oriented rather than synthesis-oriented. US11827830B2 uses a copolymer containing 15 to 50 mass % nitrile-group structural units and 40 to <85 mass % aliphatic hydrocarbon units as a dispersant; US9461308B2 uses a water-soluble polymer with aromatic vinyl, unsaturated carboxylic acid, and crosslinkable units as an electrode binder. Neither discloses NBR synthesis conditions.
- US9976027B2 and US10344158B2 are not NBR synthesis patents in the retained passages; they instead use diene-rubber core-shell particles in epoxy compositions and chlorobutadiene copolymers in dipped articles, respectively. They show that diene-rubber and elastomer design is often optimized for downstream performance rather than acrylonitrile-rich NBR monomer control.
- US10344104B2 is an ABS/SAN compatibilizer patent that explicitly allows an acrylonitrile-based monomer unit in the compatibilizer, but its core composition is maleimide/styrene/anhydride and its focus is heat resistance and compatibility, not NBR preparation.
- Across the set, the recurring historical approach is emulsion or mass graft polymerization for ABS-type materials, while the newer approach is controlled polymerization or

application-specific binder/dispersant design; direct high-ACN NBR synthesis detail is sparse in both generations.

4. CONCLUSION

The patent evidence does not reveal a dedicated, parameter-complete high-acrylonitrile NBR synthesis recipe within the manifest, but it does identify the process families most relevant to acrylonitrile-rich elastomer design. The strongest synthesis guidance comes from emulsion/graft ABS patents and the broader controlled emulsion polymerization patent: acrylonitrile-containing systems are typically built by free-radical polymerization, often with staged monomer feed, emulsifier/initiator balance, and molecular-weight control. For a high-ACN NBR target, the most defensible evidence-based recommendation is to use a controlled emulsion or solution polymerization platform, maintain tight monomer-feed control to raise acrylonitrile incorporation, and manage molecular weight with initiator and chain-transfer strategy. The set also suggests that application-driven patents favor tuning polarity, gel content, and glass transition rather than disclosing synthesis conditions. Accordingly, the best-supported synthesis direction is a radical polymerization process with careful feed staging and molecular-weight control, while recognizing that the manifest provides only indirect support for exact high-ACN NBR composition windows.

5. REFERENCES

- US9976027B2 | Polymer fine particle-containing curable resin composition having improved ... | Not disclosed | US | 2018 | <https://patents.google.com/patent/US9976027B2/en>
- US11827830B2 | Dispersant, conductive material dispersion and slurry for electrode film | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11827830B2/en>
- US10266615B2 | Method for preparing acrylonitrile-butadiene-styrene graft copolymer and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10266615B2/en>
- US4559386A | ABS compositions and process for preparing same | Not disclosed | US | 1985 | <https://patents.google.com/patent/US4559386A/en>
- US9583768B2 | Negative electrode slurry composition, lithium ion secondary battery negative ... | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9583768B2/en>
- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US9461308B2 | Electrode for lithium ion secondary battery, lithium ion secondary battery, ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9461308B2/en>
- US10344104B2 | Copolymer for polymer blend compatibilizer and resin composition | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10344104B2/en>
- US10344158B2 | Elastomeric film-forming compositions and articles made from the elastomeric ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10344158B2/en>

- US6812307B2 | Functionalized monomers for synthesis of rubbery polymers | Not disclosed | US | 2004 | <https://patents.google.com/patent/US6812307B2/en>